

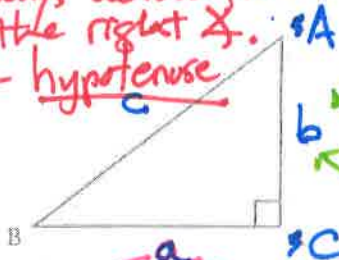
A2 NOTES T1 – RIGHT TRIANGLES AND TRIGONOMETRIC RATIOS

Trigonometry means triangle measurement. The Trig Functions relate an angle to a specific ratio of sides.

Name the sides RELATIVE to the angle!

The 3 trig ratios to know this unit are:

always across from the right $\angle$.
hypotenuse



The side opposite $\angle B$

the side adjacent to $\angle A$

The sine function: $\sin B =$

an angle
 $\frac{\text{opposite side}}{\text{hypotenuse}}$

The cosine function: $\cos B =$

$\frac{\text{adj. side}}{\text{hyp}}$

The tangent function: $\tan B =$

$\frac{\text{opp. side}}{\text{adj. side}}$

the side opposite $\angle A$ OR the side adj. to $\angle B$

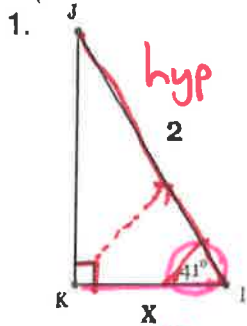
Use the acronym SOH CAH TOA to help you

$$S = \frac{O}{H} \quad C = \frac{A}{H} \quad T = \frac{O}{A}$$

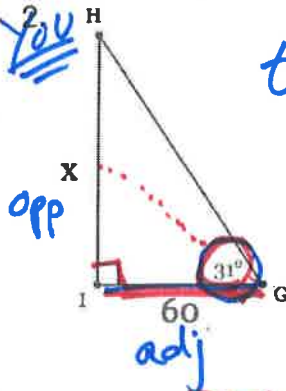
Capital Letters for $\angle$'s
corresponding lower case letters for sides opposite the angle

To Solve for a Side Length, find your angle, what you are looking for, and what is given. Choose the trig function that relates those 3 pieces.

Examples. Given each figure, solve for x . Round to the nearest tenth.

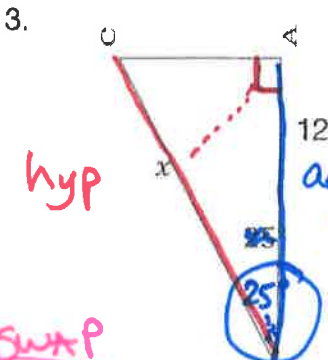


$\cos 41^\circ = \frac{x}{2}$
 $x = 2 \cos 41^\circ \approx 1.5$



$T = \frac{O}{A}$
 $\tan 31^\circ = \frac{x}{60}$
 $x = 60 \tan 31^\circ$
 $x \approx 36.1$

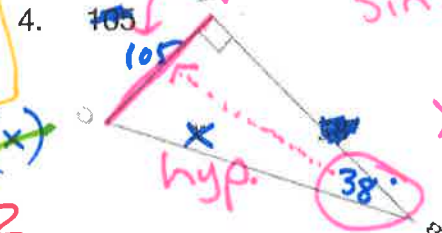
★ WATCH OUT FOR x IN THE DENOMINATOR!



$C = \frac{A}{H}$
 $\cos 25^\circ = \frac{12}{x}$

$x \cos 25^\circ = 12$
 $x = \frac{12}{\cos 25^\circ} \approx 13.2$

$\cos 25^\circ = \frac{12}{x}$



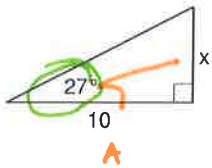
$\sin 38^\circ = \frac{105}{x}$
 $x = \frac{105}{\sin 38^\circ}$
 $x \approx 170.5$

Using Trigonometry To Find Lengths

Date _____ Period _____

Find the missing side. Round to the nearest tenth.

1)

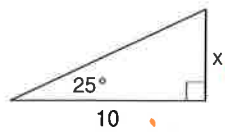


$$\tan 27^\circ = \frac{x}{10}$$

$$x = 10 \tan 27^\circ$$

$$x \approx 5.1$$

2)

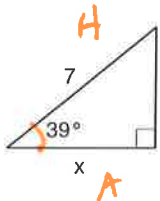


$$\tan 25^\circ = \frac{x}{10}$$

$$x = 10 \tan 25^\circ$$

$$x \approx 4.7$$

3)

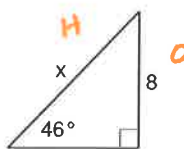


$$\cos 39^\circ = \frac{x}{7}$$

$$x = 7 \cos 39^\circ$$

$$x \approx 5.4$$

4)



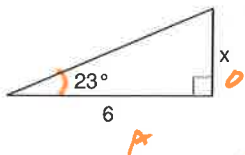
x in the denominator

$$\sin 46^\circ = \frac{8}{x} \Rightarrow \text{SWAP}$$

$$x = \frac{8}{\sin 46^\circ}$$

$$x \approx 11.1$$

5)

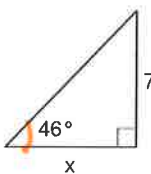


$$\tan 23^\circ = \frac{x}{6}$$

$$x = 6 \tan 23^\circ$$

$$x \approx 2.5$$

6)



$$\tan 46^\circ = \frac{7}{x} \Rightarrow \text{SWAP}$$

$$x = \frac{7}{\tan 46^\circ}$$

$$x \approx 6.8$$

7)

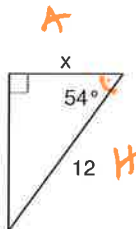


$$\tan 72^\circ = \frac{x}{20}$$

$$x = 20 \tan 72^\circ$$

$$x \approx 61.6$$

8)



$$\cos 54^\circ = \frac{x}{12}$$

$$x = 12 \cos 54^\circ$$

$$x \approx 7.1$$